

PHYS 541

Fall 2026

Office: Luter 304

Office Hours: Mon/Wed/Fri 1pm-2pm (mix of online and in person), or by appointment

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MODELING AND SIMULATION

Textbook and Software:

There is not a “required” text for this course. The problems and case studies that we work with in this course will come from a variety of sources, and I will provide you with references as we go through each section of the course.

This course is about how to solve problems on a computer. To do that, one obviously has to choose both a programming language/environment, and a development environment. There are a host of options available for each of these.

Over the years, I’ve spent a lot of time developing solutions to many different types of problems in Python, using PyCharm as the development environment, and my plan is to use Python and PyCharm, together with Jupyter Notebooks as a platform for the examples that we discuss in class. I will provide much more detail in class on how you can set up your own development environment (on whichever operating system you like).

I will make at some/most/all of these programs available to you, via GitHub, throughout the course, both as a form of lecture material and as starting points for the assignments.

Goals:

To study and understand some of the most basic and useful techniques in numerical analysis, modeling, and simulation.

To develop a suite of useful programs, methods, and libraries which you will be able to apply to a host of problems going forward.

To develop an appreciation for and understanding of some of the applications of numerical analysis in the modeling and simulation of interesting real-world processes and situations.

Topics: Numerical errors in computation, computational solutions to some simple differential equations, methods of calculating derivatives, Runge-Kutta methods, non-linear curve fitting, data science, random number generation, Monte Carlo methods, game strategy.

Homework: Assignments will be given periodically, usually once per week, and in most cases will be due approximately one week later.

Grading:	Homework	50%
	Graduate Term Project	20%
	Comprehensive Final Project	20%
	Group Leadership	10%

Homework is due by 5:00 pm on the due date. No extensions will be given, except in the case of a valid DOCUMENTED reason.

The evaluation of your performance in this course will be based entirely on the regular homework assignments and the final project. There is no possibility to do extra work for extra credit.

Significant Difference Statement:

This course is cross-listed with PHYS 441, and differs from that course in the following ways:

1. A graduate term project will be assigned to the graduate students in the course, i.e. only those students enrolled in PHYS 541, and will be more involved and complex than the typical PHYS 441 homework assignments and projects.
2. As graduate students, you are expected to take on a leadership role in group projects with students in PHYS 441.

Final grades will be assigned as follows:

A = 87-100%; A- = 80-86%
B+ = 77-79%; B = 73-76%; B- = 70-72%
C+ = 67-69%; C = 63-66%; C- = 60-62%
F < 60%

Honor Code and Cheating

As all of you know, CNU has an Honor Code, which we all agreed to by both signing and giving a pledge. The spirit of that Honor Code will be strictly observed in this course.

With that said, we need to have a longer discussion about what "violating the honor code" actually looks like in this type of course.

From my point of view, it is really very simple. I think that pretty much everything will work out well if we can agree that we will not plagiarize. The definition of plagiarism that I use is: Do not present the work of others as one's own work. In other words, don't take credit for something that someone else did. From my point of view, that does NOT mean that we cannot use the work of others, whether it is that of a classmate, or something found on the internet, or something provided by ChatGPT. It just means that we must acknowledge that we did that and give appropriate credit.

If we all follow that rule, then we should be fine!

Additional Information, provided by the CNU Administration, which you may find useful (and with which I happen to agree, in general):

**Christopher Newport University
Official Syllabi Statements
AY 2026-2027**

Academic Honesty:

Academic honesty is fundamental to the mission of a university. Cheating, plagiarism, and any other violations of the Honor Code may result in academic remedies such as a failing grade on the assignment or the course (depending on the severity of the offense). Violations will also be reported to the Center for Honor Enrichment and Community Standards (CHECS).

The Christopher Newport University Honor Code:

“On my honor, I will maintain the highest standards of honesty, integrity and personal responsibility. This means I will not lie, cheat or steal, and as a member of this academic community, I am committed to creating an environment of respect and mutual trust.”

University Statement on Inclusion and Belonging:

The Christopher Newport University community engages and respects different viewpoints, understands the cultural and structural context in which those viewpoints emerge, and questions the development of our own perspectives and values, as these are among the fundamental tenets of a liberal arts education.

Accordingly, we affirm our commitment to a campus culture that embraces the full spectrum of human attributes, perspectives, and disciplines and offers every member of the University the opportunity to become their best self.

Understanding and respecting differences can best develop in a community where members learn, live, work, and serve among individuals with diverse worldviews, identities, and values. We are dedicated to upholding the dignity and worth of all members of this academic community, such that all may engage effectively and compassionately in a pluralistic society.

If you have specific questions, suggestions, or concerns regarding inclusion on campus please contact belonging@cnu.edu.

Disabilities/Accessibility:

In order for a student to receive an accommodation due to a disability, that disability must be on record in the Office of Student Affairs, 3rd Floor, David Student Union (DSU). If you have a diagnosed disability, please contact Accessibility & Care Team Support in Student Affairs (594-7160) or email careteam@cnu.edu to discuss your needs. You may visit <https://cnu.edu/life/acts/accessibility/> to request accommodations or view additional information.

In order to provide access to educational programs in the best manner possible, students with documented disabilities should notify the instructor at least seven days prior to the

point at which they require an accommodation (the first day of class is recommended), in private, if an accommodation is needed. The instructor will provide students with disabilities with the reasonable accommodations approved and directed by the Office of Accessibility and Care Team Support. Work completed before the student notifies the instructor of his/her disability may be counted toward the final grade at the sole discretion of the instructor.

Success:

I want you to succeed in this course and at Christopher Newport. I encourage you to contact me during office hours or schedule an appointment to discuss course content or ask questions. If I become concerned about your course performance, attendance, engagement, or well-being, I will contact you first. I also may submit a referral through our Captains Care Program. The referral will be received by Student Affairs, the Center for Academic Success, and other departments, when appropriate. If you are an athlete, the Director of Student-Athlete Transition and Support will be notified. Someone from Student Affairs or the Center for Academic Success will contact you to set up a plan for success and connect you with the appropriate resources. Please remember that this is a means for me to support you and help foster your success at Christopher Newport.

Academic Support:

The Center for Academic Success (CAS) provides numerous free resources to help students succeed in their courses. These resources include individual peer mentoring, peer tutoring, and group/test review sessions. Academic Success Coordinators can meet with you regularly to discuss your plan to complete your coursework while balancing your extracurricular activities. They can help you develop individualized study skills, create time-management plans, and techniques to prepare for exams and other assignments. The Center also houses the Alice F. Randall Writing Center. Writing consultants can help you at any stage of the writing process, from invention to development of ideas to polishing a final draft.

You may contact the Center for Academic Success to request a tutor, meet with a peer mentor or writing consultant, or make an appointment to talk with a staff member about study skills and strategies. The Center is located in Christopher Newport Hall, first floor, room 124. You may email academicsuccess@cnu.edu, call (757) 594-7684, or visit <https://cnu.edu/center-for-academic-success>

Public Health: The university will provide guidance on public health issues and students will be expected to comply with university protocols.

Course Materials:

All content created and assembled by the faculty member and used in this course is to be considered intellectual property owned by the faculty member and Christopher Newport University. It is provided solely for the private use of the students currently enrolled in this course. To ensure the free and open discussion of ideas, students may not make any original course content, including but not limited to lectures, discussions, videos, handouts, and/or activities, available to anyone not currently enrolled in the course

without the instructor's advance written permission. This means that students may not record, download, screenshot, or in any way copy original course material for the purpose of distribution beyond this course. A violation may be considered theft. It is the student's responsibility to protect course materials when accessing them outside the physical classroom.